

```
1. from statsmodels.tsa.stattools import adfuller
2.
3. # Test price series for stationarity
4. price_result = adfuller(apple['Adj Close'].dropna())
5. print('ADF Statistic (Price):', price_result[0])
6. print('p-value (Price):', price_result[1])
7.
8. # Test returns series for stationarity
9. returns_result = adfuller(apple['Returns'].dropna())
10. print('ADF Statistic (Returns):', returns_result[0])
11. print('p-value (Returns):', returns_result[1])
12.
13. Typically, you'll see results like:
14. ADF Statistic (Price): -1.352
15. p-value (Price): 0.605
16. ADF Statistic (Returns): -15.789
17. p-value (Returns): 0.000
18.
```

The p-value tells us whether to reject the hypothesis of non-stationarity. A small p-value (typically <0.05) indicates the series is likely stationary. As expected, the price series is non-stationary (p-value > 0.05), while the returns series is stationary (p-value near zero).

Why does this matter? Non-stationary data can lead to spurious correlations and unreliable models. Many trading strategies have failed because they were built on non-stationary data without proper transformations.

?? **Fun Fact:** In 1991, economists David Hendry and Neil Ericsson demonstrated that you could find a nearly perfect correlation between inflation in the UK and cumulative rainfall—a classic example of spurious correlation in non-stationary data. Similarly, in trading, many apparent patterns in price charts disappear once you account for non-stationarity. When Long-Term Capital Management (LTCM) collapsed in 1998 after losing \$4.6 billion, one factor was their reliance on statistical relationships that seemed stable but broke down under stress. Their models hadn't properly accounted for the non-stationary nature of market volatility, a